# The textbook inequality is green. On a tissue of length π no admissible mode is in the band; at 2π one is.

Declared model: J = [[1, -2], [2, -3]], diffusivities D = (1, 10), 1-D Neumann boundaries; det(J − qD) = (5q-1)(2q-1), so the continuum band is 1/5 < q < 1/2. Admissible modes are q = (nπ/L)².

| LENGTH L / Π | 1/2 | 3/4 | 1 | 3/2 | 2 | 5/2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|---|---|---|---|
| linear verdict | stable | stable | stable | grows | grows | stable | grows | grows | grows | grows |

**PROVED** **COMPILED INSTANCE**

Lean 4.35.0: L = π admits no unstable mode; L = 2π admits one (n = 1, det = -1/8). Build exit 0; 3 mutants rejected. Only these two lengths are compiled.

**EXACT ARITHMETIC**

The whole strip is exact rational arithmetic. Mode n grows iff $2n^2 < (L/\pi)^2 < 5n^2$.

**OBSERVED**

**None.** We ran no experiment.

**Classical mathematics, a linear-mode statement for a declared model.** It is not a claim that a pattern forms, and not that Turing was wrong. What is new is only that a program checks it, with a witness, before a law is applied.

# A request the plain formula accepts is refused with a witness, repaired, and re-checked by an independent solver.

**REQUEST**
Target wavelength, equal diffusivities (Dv/Du = 1). 16 requests.

→

**EXACT REFUSED**
No admissible unstable mode (turing-ddi). 16 of 16.

→

**EXACT REPAIR**
Raise Dv/Du to 9 (most requests); the exact check licenses it.

→

**SIMULATED SOLVER**
Independent solver: pattern on 11 of 16 requests, mean 0.708.

The plain formula's design without repair succeeded on 0 of 16.

**Limitation first.** The algorithm is trivial. The contribution is the certificate, the refusal and the repair.

**CAVEATS, PRINTED**

One kinetics, 1-D Neumann, 3 noise seeds, no confidence interval; a model-only check, not an experiment. Near-onset repaired patterns are marginal (amplitude up to 0.18 against a 0.10 threshold). The other 5 requests did not succeed on all seeds, and repairs for 2 further refusals worked on 0. No domain-length repair was found. Across 80 nonlinear configurations the gate agreed with the solver on 80 and the plain formula on 79; linear theory named the dominant wavelength in 43/61 patterned cases.

**Ledger** unledgered (NCA worker artifacts) · artifacts/rethink_nca/repair.json · scopedesk/oracles/results/linear_vs_nonlinear.json · planning/rethink/nca.md · **Review** single-reader, not Codex-reviewed: NCA worker artifacts and a critic audit of the checker code · artifacts @ 73f52ee9

# Two compiled nodes, and a test set with exact answers that the checker is scored against.

PROVED **PROOF DEPENDENCIES**

`price · ATLAS has it; ours duplicates` →

`selection_only` → `hamilton_regression`

Conditional Hamilton regression theorem. Lean exit 0, standard axioms only, 4 witness controls, 2 changed-sign mutants rejected. A critic's code search found no Hamilton form in ATLAS; not exhaustive. Second-reader review: PENDING

**TEST SET**

Version 1.2: **127** requests, **8** families, key hash 28645a12…. **Frozen.** Codex's metadata and text check passes (14 of 14). Audit trail: a blinded payload-only audit agreed on 132 of 135 items of an earlier candidate.

**CHECKER AGAINST THE KEY**

Turing, 2-D Turing and wavelength items.

| METHOD | AGREES |
|---|---|
| **Gate with the record** | **50 / 50** |
| Plain textbook formula | 29 / 50 |
| Continuum condition only | 34 / 50 |
| Record hash altered: refuses | 50 / 50 |

**Printed caveat:** the set is trap-enriched and shares its convention document with the checker. This is not external validation. The last row shows provenance dependence only.

# The ledger: claims we held that did not survive, each with a witness and the kind of failure kept separate.

| CLAIM WE HELD | KIND | WITNESS |
|---|---|---|
| Turing needs a diffusion ratio of at least 10 | **counterexample** | a Lean-checked linear model with diffusivity ratio 1.1 has a growing mode (determinant -21/25000) |
| Pinned-end liquid bridge unstable beyond L > πr (our benchmark key) | **convention defect** | the n=1 mode violates the fixed-volume constraint; caught before any agent ran |
| Thompson's scaling refuted by a pooled regression | **mis-operationalised** | a pooled OLS tests mean allometry, not the buckling limit |
| Benchmark ids were label-neutral | **convention defect** | ids encoded generator order: a tree on id suffixes alone scored 58/96 against 43/96 for the majority answer |
| Our Price node as a novel formalization | **prior art** | atlas-lean already proves Price and Fisher; ours was a duplicate |

**20 ledger rows: 17 were caught by an independent reviewer or a measurement, 3 by the author's own check.**
There is no control arm: this compares independent audit with self-audit. It is not a measurement of collective intelligence.
Kinds: prior art 7, counterexample 1, measured 4, mis-operationalised 3, convention defect 4, process defect 1.

Also in the ledger: our theorem advertised as stability whose premise, determinant non-negativity, also holds for unstable systems (R06); a stronger statement now compiles (F10). Row R09, counted as measured, is a mismatched run (nonlinear prompts scored against linear labels), not a measured null. One team, one day: not a rate. No median is shown.

**Ledger** F04 F05 F08 F10 A01 · scopedesk/ledger/KILL-LEDGER.md and kill_ledger.json (all rows, witnesses, times and caveats) · **Review** single-reader, not Codex-reviewed: the critic's ledger (Codex corrected only R06); rows not independently approved · artifacts @ 73f52ee9

# What the collective measured: more agents did not help on the controlled tasks, and the proof pilot hit a ceiling.

**SIMULATED** **ENZYME REPLAY**

159,129 measured variants, 1,000 extra assays.




No collective advantage. One repeated panel, about 24 distinct problems.

**HEADROOM CHECK ON THE FROZEN TEST SET**

Unguarded models, no code tool: Claude Sonnet 5.5 **121/127** (95.3%), Gemini 3.8 flash **110/127** (86.6%). Checker-tool arm not run. Effective n is 8 families. This measures judgement, not derivation, and is a coarse consistency check scored by the key's author; not a model ranking. Gemini's misses sit in wavelength (7 of 14) and Greenhill (6 of 13).

**COMPILED** **PROOF-SEARCH PILOT, 19 FROZEN INTEGER-ARITHMETIC LEMMAS**

Same model, 3 replicates, 30 calls per run: every arm certified 19/19 in every run, a ceiling. Library reuse is real but saved nothing: median calls none 21, private 22, sequential 20, shared 21; median tokens 72k, 80k, 78k, 80k. Deterministic tactics certified 4/19. All 228 recorded proofs re-certified after the checker was repaired, a repair made after the outcomes. Call-matched, not compute-matched; arithmetic surrogates, not biology.

**Ledger** B02 B03 P02 P03 · artifacts/core/replay/table_primary.csv · tasks/results/guilds.json · artifacts/bench_headroom/ · lean_commons/PREREG.md (Addendum D) · **Review** single-reader, not Codex-reviewed: the guilds results, scored by the key’s author; Codex’s review of those statements is advisory only · artifacts @ 73f52ee9

# Reusable now: a checker that returns a witness. Outside use: none observed yet.

### NOW

The checker, repair enumerator, test set and ledger are in this repository. A fresh-clone acceptance run passed (exit 0, 56.6 s) at an earlier commit; a receipt for the final commit is PENDING

### NEXT, A PROPOSAL

A pilot on a modelling team's recent reuse decisions. Delphi Frontier, nothing deployed: one page per open problem showing what we know, what's next and who is working on it. Misprediction map over a large ensemble: PENDING

### CREDIT, AND WHAT WE DO NOT CLAIM

ATLAS (facebookresearch/atlas-lean), Prove2Me (arXiv 2608.28433), Rashid et al. 2017, lean-bio, crnt-lean, Ripple (arXiv 2607.13531). No first library, no universal threshold, and no claim that a named law is wrong.

### SCORECARD

21 records: 6 linked to scope-check benchmark families, 1 related formal lemma (target unestablished), 2 empirical/framework, 12 without a corresponding check. Single-reader statuses; Codex reviewed the corrected descriptions. The 12 are a backlog, not open problems, and the 6 are not 6 false laws.

## Make the assumptions behind a biological-model decision executable and auditable.

**Ledger** A01 A02 P01 P07 P08 · scopedesk/scorecard/ (critic, v0) · scopedesk/oracles/results/accept_receipt_fresh_clone.json · planning/rethink/PRIOR-ART-VERDICT.md · **Review** single-reader, not Codex-reviewed: scorecard statuses (Codex reviewed the corrected descriptions earlier; no re-review) · artifacts @ 73f52ee9